# Uncertainty Quantification in Machine Learning

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# Chapter 1

# Assignment 1 Done 2026 September 24 Due 2026 September 29

**Remark** All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

## Exercise 1

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$ and a given model $\hat{f} : \mathcal{X} \to \mathbb{R}$. Construct a score function $s(x, y)$ that would yield intervals of the form $\mathcal{C}(x) = [\hat{f}(x) - \gamma, \hat{f}(x) + 2\gamma]$ for some $\gamma > 0$.

**Solution** Put

$$s(x,y) = \begin{cases} 2\left|\hat{f}(x) - y\right| & y < \hat{f}(x) \\ \left|\hat{f}(x) - y\right| & y \geqslant \hat{f}(x). \end{cases}$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leqslant \hat{q}\}$, we will have the desired interval with $\gamma = \frac{\hat{q}}{2}$. ////

## Exercise 2

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}^d$ and a given model $\hat{f} : \mathcal{X} \to \mathbb{R}^d$ whose coordinates are $\hat{f}(x) = (\hat{f}_1(x), \ldots, \hat{f}_d(x))$. Construct a score function $s(x, y)$ that would yield intervals of the form

$$\mathcal{C}(x) = \prod_{j=1}^{d} [\hat{f}_j(x) - c, \hat{f}_j(x) + c]$$

for a $c > 0$.

**Solution** Put

$$s(x,y) = \max_{j=1,\ldots,d} \left|\hat{f}_j(x) - y_j\right|.$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leqslant \hat{q}\}$, we will have the desired interval with $c = \hat{q}$. ////

# Exercise 3

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$, two given models $\hat{f}_i : \mathcal{X} \to \mathbb{R}$ for $i = 1, 2$, and the score function

$$s(x, y) = \min_{i=1,2} \left| y - \hat{f}_i(x) \right|.$$

Derive an explicit expression for $\mathcal{C}(x)$ in terms of $\hat{f}_1(x)$, $\hat{f}_2(x)$, and the quantile $\hat{q}$ of the scores on the validation set. What is the shape of the prediction when $\hat{q}$ is small or when it is large? Briefly explain why this might be attractive if the conditional distribution of $Y$ given $X = x$ is bimodal.

**Solution** By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leqslant \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at $x$, that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of $Y$ given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////

# Exercise 4

Fix $n > 2$. Construct a random vector $(X_1, \ldots, X_n)$ defined on some probability space such that $(X_i, X_j)$ is exchangeable for all $i \neq j$ but $(X_1, \ldots, X_n)$ is not exchangeable.

**Solution** Let $(X, Y)$ be a jointly normal random vector. It is exchangeable if and only if $\mathbb{E}[X] = \mathbb{E}[Y]$ and $\text{Var}(X) = \text{Var}(Y)$. With this in mind, let $(X_1, \ldots, X_n)$ be a jointly normal random vector such that $\mathbb{E}[X_i] = \mathbb{E}[X_j]$ and $\text{Var}(X_i) = \text{Var}(X_j)$ for all $i \neq j$ but $\text{Cov}(X_1, X_2) \neq \text{Cov}(X_1, X_3)$. Then $(X_i, X_j)$ is exchangeable for all $i \neq j$ but $(X_1, \ldots, X_n)$ is not exchangeable. For a concrete example, say $\mathbb{E}[X_i] = 0$ and $\text{Var}(X_i) = 1$ for all $i$, $\text{Cov}(X_1, X_2) = -1$, and all other covariances are zero. ////

# Exercise 5

Fix $n \geqslant 1$, $\tau \in (0, 1)$. For any vector $v \in \mathbb{R}^n$, define $\text{Quantile}(v; \tau) = v_{(\lceil \tau n \rceil)}$, where $v_{(1)} \leqslant v_{(2)} \leqslant \ldots \leqslant v_{(n)}$ are the order statistics of $v$. Under what condition is it true that, for all $v \in \mathbb{R}^n$, $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$?

**Solution** The order statistics of $-v$ are given by $(-v)_{(i)} = -v_{(n-i+1)}$ for $i = 1, \ldots, n$. Therefore, $\text{Quantile}(v; 1-\tau) = -\text{Quantile}(-v; \tau)$ holds for a vector $v \in \mathbb{R}^n$ if and only if

$$v_{(\lceil (1-\tau)n \rceil)} = -((-v)_{(\lceil \tau n \rceil)}) = -(-v_{(n - \lceil \tau n \rceil + 1)}) = v_{(n - \lceil \tau n \rceil + 1)}.$$

To make this work for all $v \in \mathbb{R}^n$, the condition we must pose is that

$$\lceil (1 - \tau)n \rceil = n - \lceil \tau n \rceil + 1.$$

This can happen for instance when $\tau = 1/2$ and $n$ is odd. ////

# Exercise 6

For $z \in \mathbb{R}^n$, write $\hat{F}_z(v) = \frac{1}{n}\sum_{i=1}^n \chi_{\{z_i \leq v\}}$ for the empirical cumulative distribution function of $z$ with $\hat{F}_z(-\infty) = 0$ and $\hat{F}_z(+\infty) = 1$. Write

$$\text{Quantile}(z;\tau) = \inf\left\{v \in \mathbb{R} \mid \hat{F}_z(v) \geqslant \tau\right\}$$

for the empirical $\tau$-quantile of $z$ for any $\tau \in [0,1]$. Consequently, $\text{Quantile}(z;0) = -\infty$.

Prove that, for any $z \in \mathbb{R}^n$, the following statements are true:

1. $\hat{F}_z(v) = \sup\{\tau \in [0,1] \mid \text{Quantile}(z;\tau) \leqslant v\}$ for all $v \in \mathbb{R}$.
2. $\text{Quantile}(z; \hat{F}_z(v)) \leqslant v$ for all $v \in \mathbb{R}$.
3. $\hat{F}_z(\text{Quantile}(z;\tau)) \geqslant \tau$ for all $\tau \in [0,1]$.
4. If $z$ has no repeated values, then $\hat{F}_z(\text{Quantile}(z;\tau)) = \frac{\lceil n\tau \rceil}{n}$ for all $\tau \in [0,1]$.

**Solution** Let $z \in \mathbb{R}^n$ be given and fixed. Let us inrtoduce a notation $S_\tau$ for the set of all $v \in \mathbb{R}$ such that $\hat{F}_z(v) \geqslant \tau$. Then, by definition, $\text{Quantile}(z;\tau) = \inf S_\tau$. All four statements depend on the following lemma:

**Lemma 1.** If $v \in \mathbb{R}$ and $\tau \in [0,1]$, then $\text{Quantile}(z;\tau) \leqslant v$ if and only if $\tau \leqslant \hat{F}_z(v)$.

*Proof.* If $\tau \leqslant \hat{F}_z(v)$, then $v \in S_\tau$ and $v \geqslant \inf S_\tau = \text{Quantile}(z;\tau)$.

Let us focus on the converse. Suppose that $\text{Quantile}(z;\tau) \leqslant v$. For simplicity, put $q = \text{Quantile}(z;\tau)$. If $\tau = 0$, then $\hat{F}_z(v) \geqslant 0 = \tau$ holds trivially. Assume from now on that $\tau > 0$. Then $q$ is finite number. Choose $\delta > 0$ small enough such that no points $z_i$ lies between $q$ and $q + \delta$, which is always doable because $z$ has only finitely many points. Since $q$ is the infimum, choose $s \in S_\tau$ such that $q \leqslant s < q + \delta$. Then, we arrive at

$$\hat{F}_z(v) \geqslant \hat{F}_z(q) = \hat{F}_z(s) \geqslant \tau$$

by applying in the first inequality the monotonicity of $\hat{F}_z$ and in the second equality the fact that no points $z_i$ lies between $q$ and $q + \delta$. The lemma is proved. □

We now prove the four statements. To prove the first statement, let $v \in \mathbb{R}$ be given and we have, by the lemma:

$$\{\tau \in [0,1] \mid \text{Quantile}(z;\tau) \leqslant v\} = \left\{\tau \in [0,1] \mid \tau \leqslant \hat{F}_z(v)\right\} = [0, \hat{F}_z(v)]$$

and the suprenum of the interval is just $\hat{F}_z(v)$. The second and the third statements are already included in the lemma. The fourth statement is trivial given the special assumption that $z$ has no repeated values.////

# Exercise 7

Consider the conformal prediction with $\mathcal{Y} = \mathbb{R}$ with a given model $\hat{f} : \mathcal{X} \to \mathcal{Y}$ and calibration points $(X_1, Y_1)$,..., $(X_n, Y_n)$. Put

$$\hat{q}_L = \text{Quantile}\left(\left\{\hat{f}(X_i) - Y_i\right\}_{i=1}^n ; \left(1 - \frac{\alpha}{2}\right)\left(1 + \frac{1}{n}\right)\right),$$

$$\hat{q}_R = \text{Quantile}\left(Y_i - \left\{\hat{f}(X_i)\right\}_{i=1}^n ; \left(1 - \frac{\alpha}{2}\right)\left(1 + \frac{1}{n}\right)\right),$$

and

$$\mathcal{C}(X_{n+1}) = \left[\hat{f}(X_{n+1}) - \hat{q}_L, \hat{f}(X_{n+1}) + \hat{q}_R\right].$$

Prove that

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \geqslant 1-\alpha$$

by showing both

$$P\left\{Y_{n+1} < \hat{f}(X_{n+1}) - \hat{q}_L\right\} \leqslant \frac{\alpha}{2}$$

and

$$P\left\{Y_{n+1} > \hat{f}(X_{n+1}) + \hat{q}_R\right\} \leqslant \frac{\alpha}{2},$$

assuming that $(X_1, Y_1), \ldots, (X_{n+1}, Y_{n+1})$ is exchangeable.

**Solution** We have

$$P\left\{Y_{n+1} < \hat{f}(X_{n+1}) - \hat{q}_L\right\} = P\left\{\hat{f}(X_{n+1}) - Y_{n+1} > \hat{q}_L\right\} = 1 - P\left\{\hat{f}(X_{n+1}) - Y_{n+1} \leqslant \hat{q}_L\right\} \leqslant 1 - \left(1 - \frac{\alpha}{2}\right) = \frac{\alpha}{2}$$

where the last inequality uses Theorem 3.2 in the textbook with score $S_i = \hat{f}(X_i) - Y_i$. This is legitimate because the data is assumed to be exchangeable and the score is trivially symmetric in case of split conformal prediction. The right hand side inequality is proved in exactly the same way but with score $S_i = Y_i - \hat{f}(X_i)$. Combine the two, we have the desired marginal coverage. The proof is complete. ////

# Exercise 8

Consider the full conformal prediction with $(X_1, Y_1), \cdots, (X_{n+1}, Y_{n+1})$ being exchangeable and a symmetric score function $s$, prove that

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \leqslant 1 - \alpha + \mathbb{E}\left[\frac{1}{n+1}\sum_{i=1}^{n+1} \chi_{\{S_i = \hat{q}\}}\right]$$

where

$$\hat{q} = \text{Quantile}\left(S_1, \ldots, S_{n+1}; 1-\alpha\right).$$

Give an example showing that this bound is tighter than

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \leqslant \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1} + \epsilon_{\text{tie}}$$

where $\epsilon_{\text{tie}}$ is the probability of $S_{n+1}$ being equal to some $S_j$ for $j \in \{1, \ldots, n\}$.

**Solution** As already discussed in the lecture and in the textbook, $Y_{n+1} \in \mathcal{C}(X_{n+1})$ if and only if $S_{n+1} \leqslant \hat{q}$ and so

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} = P\left\{S_{n+1} \leqslant \hat{q}\right\} = P\left\{S_{n+1} < \hat{q}\right\} + P\left\{S_{n+1} = \hat{q}\right\}.$$

Given that $(S_1, \ldots, S_n, S_{n+1})$ is exchangeable, $P\left\{S_{n+1} = \hat{q}\right\} = P\left\{S_i = \hat{q}\right\}$ for all $i \in \{1, \ldots, n\}$ and

$$P\left\{S_{n+1} = \hat{q}\right\} = \frac{1}{n+1}\sum_{i=1}^{n+1} P\left\{S_i = \hat{q}\right\} = \mathbb{E}\left[\frac{1}{n+1}\sum_{i=1}^{n+1} \chi_{\{S_i = \hat{q}\}}\right].$$

The only thing left is to show that $P\{S_{n+1} < \hat{q}\} \leqslant 1-\alpha$. To do so, first observe that $S_{n+1} < \hat{q}$ if and only if $S_{n+1} < S_{(k)}$, where $k = \lceil (1-\alpha)(n+1) \rceil$. Thus:

$$P\{S_{n+1} < \hat{q}\} = P\left\{S_{n+1} < S_{(k)}\right\} \leqslant \frac{\lceil (1-\alpha)(n+1) \rceil - 1}{n+1} = \frac{\lfloor (1-\alpha)(n+1) \rfloor}{n+1} \leqslant \frac{(1-\alpha)(n+1)}{n+1} = 1-\alpha.$$

This completes the proof.

Finally, in case when the random variables $S_i$'s are all continuous random variables, $\epsilon_{\text{tie}} = P\{S_{n+1} = \hat{q}\} = 0$ and evidently the bound that is worked out in this exercise is better. In case when the random variables $S_i$'s are discrete and can only take very limited values, it may happen that $S_1, \ldots, S_n$ has taken all possible values and $\epsilon_{\text{tie}} = 1$ so that the bound worked out here is really the only available bound. ////

# Exercise 9

Given data set $(X_1, Y_1), \ldots, (X_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$ where $\mathcal{Y} = \mathbb{R}$, consider split conformal prediction with two given models $\hat{f}_i : \mathcal{X} \to \mathbb{R}$, the score functions

$$s_i(x, y) = \left| y - \hat{f}_i(x) \right|,$$

and two split conformal prediction sets

$$\mathcal{C}_i(X_{n+1}) = \{y \in \mathcal{Y} \mid s_i(X_{n+1}, y) \leqslant \hat{q}_i\},$$

where

$$\hat{q}_i = \text{quantile}\left\{ \{s_i(X_j, Y_j)\}_{j=1}^n ; (1-\alpha)(1+\frac{1}{n}) \right\}.$$

Put

$$L_i = \frac{1}{n} \sum_{j=1}^{n} (Y_j - \hat{f}_i(X_j))^2$$

and $I = \arg\min_{i \in \{1,2\}} L_i$. Explain why

$$P\{Y_{n+1} \in \mathcal{C}_I(X_{n+1})\} \geqslant 1-\alpha$$

may not hold even if we assume that $(X_1, Y_1), \ldots, (X_{n+1}, Y_{n+1})$ are exchangeable.

**Solution** In one sentence, the reason is that even if we assume that the three random vectors $(X_1, Y_1), \ldots, (X_{n+1}, Y_{n+1})$, $s_1(X_1, Y_1), \ldots, s_1(X_{n+1}, Y_{n+1})$, and $s_2(X_1, Y_1), \ldots, s_2(X_{n+1}, Y_{n+1})$ are all exchangeable, in general it is still not true that $s_I(X_1, Y_1), \ldots, s_I(X_{n+1}, Y_{n+1})$ is exchangeable and thus the theorem guaranteeing the marginal coverage does not in general apply.

To show that things can indeed fail, we give a simple counterexample. Put $\mathcal{X} = \mathbb{R}^2$, $n = 1$, $\alpha = \frac{1}{2}$, $X_j = (X_j^{(1)}, X_j^{(2)})$, $X_j$ follows $N_2(0, I_2)$ distribution, $Y_j = X_j^{(1)} + X_j^{(2)}$, $\hat{f}_1(x) = x^{(1)}$ and $\hat{f}_2(x) = x^{(2)}$. It is clear that, for $i = 1, 2$, $\hat{q}_i = s_i(X_1, Y_1)$, $s_i(X_j, Y_j) = \left| X_j^{(2-i)} \right|$, and $L_i = (X_1^{(2-i)})^2$. Thus,

$$\begin{aligned}
&P\{s_I(X_2, Y_2) \leqslant \min(s_1(X_1, Y_1), s_2(X_1, Y_1))\} \\
&= P\{s_1(X_2, Y_2) \leqslant \min(s_1(X_1, Y_1), s_2(X_1, Y_1)), L_1 < L_2\} \\
&+ P\{s_2(X_2, Y_2) \leqslant \min(s_1(X_1, Y_1), s_2(X_1, Y_1)), L_1 > L_2\} \\
&= P\left\{ \left|X_2^{(2)}\right| \leqslant \left|X_1^{(2)}\right| \leqslant \left|X_1^{(1)}\right| \right\} + P\left\{ \left|X_2^{(1)}\right| \leqslant \left|X_1^{(1)}\right| \leqslant \left|X_1^{(2)}\right| \right\} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}
\end{aligned}$$

and

$$P\left\{Y_2 \in \mathcal{C}_I(X_2)\right\} = P\left\{s_I(X_2, Y_2) \leqslant \min(s_1(X_1, Y_1), s_2(X_1, Y_1))\right\} = \frac{1}{3} < \frac{1}{2} = 1 - \alpha.$$

This completes the counterexample. ////

# Chapter 2

# Assignment 2 Done 2026 October 16 Due 2026 October 18

**Remark** All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

## Exercise 1

Consider split conformal prediction with data set $(X_1, Y_1), \cdots, (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$ and the conformal risk control with given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all $i$ for simplicity. Instead of what is in Chapter 10 of the book, we define, for all $\lambda \geqslant 0$:

$$\mathcal{C}_\lambda(x) = \{y \in \mathcal{Y} \mid s(x, y) < \lambda\}$$

and

$$L(y, \mathcal{C}_\lambda(x)) = \chi_{\{y \notin \mathcal{C}_\lambda(x)\}}.$$

Show that, with this definition, $L(y, \mathcal{C}_\lambda(x))$ is no longer always right continuous for every fixed $(x, y) \in \mathcal{X} \times \mathcal{Y}$. Next, given $\alpha \in (0, 1)$, give a concrete expression for

$$\hat{\lambda} = \inf \left\{ \lambda \geqslant 0 \mid \hat{R}(\lambda) \leqslant \alpha - \frac{\alpha - 1}{n} \right\}$$

in terms of the data, where

$$\hat{R}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} L(Y_i, \mathcal{C}_\lambda(X_i)).$$

Show that, if we assume that $(S_1, \ldots, S_{n+1})$ is exchangeable and that the $S_i$'s almost surely has no tie, then

$$P\left\{Y_{n+1} \in \mathcal{C}_{\hat{\lambda}}(X_{n+1})\right\} \geqslant 1 - \alpha.$$

Provide a counterexample to show that this need not hold if there are ties in $S_i$'s with positive probability, even if they are exchangeable.

**Solution** Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$ be given. If $\{\lambda_m\}_{m=1}^{\infty}$ is any decreasing sequence such that $\lambda_m > s(x, y)$ for every $m$ and $\lambda_m \to s(x, y)$ as $m \to \infty$, then $L(y, \mathcal{C}_{\lambda_m}(x)) = 1$ for all $m$ but $L(y, \mathcal{C}_{s(x,y)}(x)) = 0$, proving that $L(y, \mathcal{C}_\lambda(x))$ is not always right continuous.

We move to compute $\hat{\lambda}$. The definition of $\hat{R}(\lambda)$ and simple arithmetic rearrangement shows that, for any $\lambda \geqslant 0$,

$$\hat{R}(\lambda) \leqslant \alpha - \frac{\alpha - 1}{n}$$

if and only if

$$\sum_{i=1}^{n} \chi_{\{S_i < \lambda\}} \geqslant (n+1)(1-\alpha).$$

Thus, it is obvious that

$$\hat{\lambda} = S_{(\lceil (n+1)(1-\alpha) \rceil)}.$$

Here we have used order statistics $S_{(1)} \leqslant \cdots \leqslant S_{(n)}$ for the first $n$ values of $S_1, \ldots, S_n$. Consequently:

$$C_{\hat{\lambda}}(X_{n+1}) = \left\{ y \in \mathcal{Y} \mid s(X_{n+1}, y) < S_{(\lceil (n+1)(1-\alpha) \rceil)} \right\}.$$

To prove coverage, let $\alpha \in (0,1)$ be given. Assume first that that $(S_1, \ldots, S_{n+1})$ is exchangeable and there is almost surely no tie in the $S_i$'s. Then

$$P\left\{ Y_{n+1} \in C_{\hat{\lambda}}(X_{n+1}) \right\} = P\left\{ S_{n+1} < S_{(\lceil (n+1)(1-\alpha) \rceil)} \right\} = \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geqslant 1 - \alpha,$$

where the second equality uses both assumptions: when $S_{(1)} < \cdots < S_{(n)}$, inserting a new exchangeable point $S_{n+1}$ to get exactly right before $\lceil (n+1)(1-\alpha) \rceil$th place has the probability $\frac{\lceil (n+1)(1-\alpha) \rceil}{n+1}$. If we remove the assumption that there are no ties, then everything can go wrong. For example, just setting $s \equiv 0$. Then $\hat{\lambda} = 0$ and $C_{\hat{\lambda}}(x) = \emptyset$ for all $x$, clearly not yielding any good coverage probability. ////

# Exercise 2

Consider split conformal prediction with data set $(X_1, Y_1), \cdots, (X_{n+2}, Y_{n+2}) \in \mathcal{X} \times \mathcal{Y}$ with given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all $i$ for simplicity. For $j \in \{1, 2\}$, the conformal $p$-value is the random variable defined by

$$p_j = \frac{1 + \sum_{i=1}^{n} \chi_{\{S_{n+j} \leqslant S_i\}}}{n+1}.$$

Given $\alpha \in (0,1)$, let $\mathcal{C}$ be the conformal interval calibrated only on the first $n$ observations. In other words, for $j \in \{1, 2\}$, the event $\{Y_{n+j} \in \mathcal{C}(X_{n+j})\}$ equals to $\{p_j < \alpha\}$. Assuming that $((X_1, Y_1), \cdots, (X_{n+2}, Y_{n+2}))$ is exchangeable, $s$ is a symmetric score function, and that the random vector $(p_1, p_2)$ satisfies the positive regression dependence on subsets (PRDS) assumption given in definition 10.3 of the textbook, prove that

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1}),\, Y_{n+2} \in \mathcal{C}(X_{n+2})\} \geqslant (1-\alpha)^2$$

without explicitly computing the probability on the left hand side of the inequality.

**Solution** First of all, Theorem 3.2 in the textbook already guarantees that, for $j \in \{1, 2\}$:

$$P\{Y_{n+j} \in \mathcal{C}(X_{n+j})\} \geqslant 1 - \alpha.$$

We focus on showing that

$$P\left\{ Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1}) \right\} \geqslant 1 - \alpha.$$

In order to make use of the PRDS condition, define a function $g$ on the support of the discrete random variable $p_1$ by setting, for all $t$, that

$$g(t) = P\left(\{p_2 > \alpha\} \mid p_1 = t\right).$$

Note the following facts:

1. $g$ is monotonically increasing because $g(t) = P\left(\{(p_1, p_2) \in A\} \mid p_1 = t\right)$ with the set $A$ defined by all $(u, v) \in \mathbb{R}^2$ with $v < \alpha$, who is an increasing set.
2. The average value of $g(p_1)$ on the set $\{p_1 > \alpha\}$ is the desired conditional probability. In symbols:

$$\frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\}.$$

3. The expectation of $g(p_1)$ is the marginal covergae of $(X_{n+2}, Y_{n+2})$. In symbols:

$$\mathbb{E}\left[g(p_1)\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2})\right\}.$$

4. The average value of $g(p_1)$ on the set $\{p_1 > \alpha\}$ is greater or equal to the average value of $g(p_1)$ on the set $\{p_1 \leqslant \alpha\}$ because $g$ is monotonically increasing. That is:

$$\frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] \geqslant \frac{1}{P\{p_1 \leqslant \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 \leqslant \alpha\}\right]$$

The proof is complete once we combine the last three facts with the marginal coverage on $(X_{n+2}, Y_{n+2})$:

$$\begin{aligned}
&1 - \alpha \geqslant P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2})\right\} = \mathbb{E}\left[g(p_1)\right] \\
&= \frac{P\{p_1 > \alpha\}}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] + \frac{P\{p_1 \leqslant \alpha\}}{P\{p_1 \leqslant \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 \leqslant \alpha\}\right] \\
&\geqslant (P\{p_1 > \alpha\} + P\{p_1 \leqslant \alpha\})\frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] \\
&= \frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\}.
\end{aligned}$$

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# Exercise 3

Suppose $(X_1, Y_1), \ldots, (X_{n+2}, Y_{n+2}) \in \mathcal{X} \times \mathcal{Y}$ are exchangeable. After observing $X_{n+1}$ and $X_{n+2}$, select $I \in \{1, 2\}$, where $I$ may depend on both $(X_{n+1}, X_{n+2})$. Given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, define a (full) conformal prediction set $\mathcal{C}(X_{n+I})$ such that

$$P\{Y_{n+I} \in \mathcal{C}(X_{n+I})\} \geqslant 1 - \alpha.$$

**Solution** The solution is to use a Benferroni correction for the two test points.

Specifically, put $S_i = s(X_i, Y_i)$ and let $S_{(1)} \leqslant \cdots \leqslant S_{(n)}$ be the order statistics for $S_1, \ldots, S_n$. Define

$$k = \left\lceil (n+1)\left(1 - \frac{\alpha}{2}\right)\right\rceil,$$

$$\hat{q} = \begin{cases} S_{(k)}, & k \leqslant n, \\ \infty, & k = n+1, \end{cases}$$

and

$$\mathcal{C}(X_{n+I}) = \{y \in \mathcal{Y} \mid s(X_{n+I}, y) \leqslant \hat{q}\}.$$

Standard results (for example, for full conformal prediction, Theorem 3.2 in the textbook) show that, for $i \in \{1, 2\}$, we have

$$P\{Y_{n+i} \notin \mathcal{C}(X_{n+i})\} \leqslant \frac{\alpha}{2}.$$

Thus:

$$P\{Y_{n+I} \notin \mathcal{C}(X_{n+I})\} = P\left(\bigcup_{i=1}^{2} \{Y_{n+i} \notin \mathcal{C}(X_{n+i})\}\right) \leqslant 2 \times \frac{\alpha}{2} = \alpha.$$

This completes the proof. ////

# Exercise 4

Consider split conformal prediction with calibration data set $(X_1, Y_1), \cdots, (X_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$ at given level $\alpha \in (0, 1)$ and given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all $i$ for simplicity. We look at the aggregating prediction sets by the majority vote $\mathcal{C}^{\text{mv}}(X_{n+1})$ of two conformal algorithms $\mathcal{C}_1$ and $\mathcal{C}_2$, where $\mathcal{C}_1$ uses $s$ as score and $\mathcal{C}_2$ uses $-s$ as score. Give an explicit formula of $\mathcal{C}^{\text{mv}}(X_{n+1})$ and its marginal coverage level, assuming exchangeability of the data set, symmetry of $s$, and almost surely no ties in $S_i$'s.

**Solution** We first compute the expression of each conformal set. Let us use the order statistics $S_{(1)} < \cdots < S_{(n)}$ of $S_1, \cdots, S_n$. Put

$$k = \lceil (n+1)(1-\alpha) \rceil.$$

We discuss by cases of whether $k = n + 1$.

First of all, in the usual case, $k < n + 1$. The rule $(-v)_{(i)} = -v_{(n-i+1)}$ for any $v \in \mathbb{R}^n$ tells us that, by definition of $\mathcal{C}_1$ and $\mathcal{C}_2$:

$$\mathcal{C}_1(x) = \left\{y \in \mathcal{Y} \mid s(x, y) \leqslant S_{(k)}\right\}$$

and

$$\mathcal{C}_2(x) = \left\{y \in \mathcal{Y} \mid s(x, y) \geqslant S_{(n+1-k)}\right\}.$$

Thus, by definition:

$$\mathcal{C}^{\text{mv}}(x) = \left\{y \in \mathcal{Y} \mid \frac{1}{2}\sum_{i=1}^{2} \chi_{y \in C_i(x)} > \frac{1}{2}\right\} = \mathcal{C}_1(x) \cap \mathcal{C}_2(x) = \left\{y \in \mathcal{Y} \mid S_{(n+1-k)} \leqslant s(x, y) \leqslant S_{(k)}\right\}.$$

Next, for the trivial case $k = n + 1$, we have $\mathcal{C}^{\text{mv}}(x) = \mathcal{C}_1(x) = \mathcal{C}_2(x) = \mathcal{Y}$. Therefore:

$$\mathcal{C}^{\text{mv}}(X_{n+1}) = \begin{cases} \{y \in \mathcal{Y} \mid S_{(n+1-k)} \leqslant s(X_{n+1}, y) \leqslant S_{(k)}\} & k = \lceil (n+1)(1-\alpha) \rceil < n+1, \\ \mathcal{Y} & k = n+1. \end{cases}$$

A simple counting of the probability of placing $S_{n+1}$ gives

$$P\left\{Y_{n+1} \in \mathcal{C}^{\text{mv}}(X_{n+1})\right\} = \frac{\max\{0, 2k-n-1\}}{n+1} = \max\left\{0, 1 - \frac{2\lfloor \alpha(n+1) \rfloor}{n+1}\right\}.$$

It just happens that this formula works for $k = n+1$ case as well. This completes the computation. ////

# Chapter 3

# Assignment 3 Done 2026 October 25 Due 2026 November 1

**Remark** All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

## Exercise 1

Given exchangeable data set $(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$, level $\alpha \in (0, 1)$, and a real valued symmetric score function $s((x, y); \mathcal{D})$ for $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and $\mathcal{D} \in (\mathcal{X} \times \mathcal{Y})^k$ for any $k$ being a data set, we carry out the full conformal prediction. Put, for any $y \in \mathcal{Y}$:

$$\mathcal{D}_{n+1}^y = \{(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, y)\}.$$

For simplicity, write $S_i^y = s((X_i, Y_i); \mathcal{D}_{n+1}^y)$ for $i \in \{1, \ldots, n\}$ and $S_{n+1}^y = s((X_{n+1}, y); \mathcal{D}_{n+1}^y)$. The standard conformal prediction procedure asks us to compare $S_{n+1}^y$ with the entire sequence $(S_1^y, \ldots, S_n^y)$.

We proceed however by choosing randomly a subset $I \subseteq \{1, \ldots, n\}$ independent of the data and compare $S_{n+1}^y$ with $(S_i^y)_{i \in I}$. More precisely, put:

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathcal{Y} \mid S_{n+1}^y \leqslant \text{Quantile}\left( (S_i^y)_{i \in I}; (1-\alpha)\left(1 + \frac{1}{|I|}\right)\right)\right\}.$$

Prove that

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \geqslant 1 - \alpha.$$

**Solution** Given that $I$ is independent of the data set, the joint distribution of $((X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, Y_{n+1}))$ remains the same after conditioning on $I$. In particular, they are still exchangeable. Therefore:

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} = \sum P\{I = J\} P\left(\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \mid I = J\right) \geqslant \sum P\{I = J\}(1-\alpha) = 1 - \alpha,$$

where the summation is taken with $J$ ranged over all subsets of $1, \ldots, n$. The second inequality is Theorem 3.2 in the textbook. ////

# Exercise 2

Fix positive integer $K$ and consider the $K$-nearest neighbor regression with $\mathcal{X}$ being any set (any metric space) and $\mathcal{Y} = \mathbb{R}$ who gives a fitted model

$$\hat{f}(x; \mathcal{D}) = \frac{1}{K} \sum_{i \in N(x, \mathcal{D})} y_i,$$

where $\mathcal{D}$ is any data set containing points from $\mathcal{X} \times \mathcal{Y}$ and $N(x, \mathcal{D})$ denotes indices of all $K$ points in $\mathcal{D}$ whose $\mathcal{X}$ component is, in some sense, closest to $x$. We can assume that there is almost surely no ties by implementing any tie breaking rule.

Now let us consider the data set

$$\mathcal{D}_n = \{(X_1, Y_1), \cdots, (X_n, Y_n)\}.$$

For any $y \in \mathcal{Y}$, put

$$\mathcal{D}_{n+1}^y = \{(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, y)\}.$$

Prove that there exists real valued functions $a(x; \mathcal{D}_n, X_{n+1})$ and $b(x; \mathcal{D}_n, X_{n+1})$ whose values depend only on $x$, $\mathcal{D}_n$, and $X_{n+1}$ such that, for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$:

$$\hat{f}(x; \mathcal{D}_{n+1}^y) = a(x; \mathcal{D}_n, X_{n+1}) + b(x; \mathcal{D}_n, X_{n+1}) y.$$

**Solution** First observe the fact that the set $N(x, \mathcal{D}_{n+1}^y)$ merely reads the set of all indices $i \in \{1, \ldots, n+1\}$ belonging to the $K$ points amongst $X_1, \ldots, X_{n+1}$ who are the closest to $x$. Therefore, although $y$ is present in the notation, $N(x, \mathcal{D}_{n+1}^y)$ depends only on $x$ and the points $X_1, \ldots, X_{n+1}$. Therefore, by simply writing

$$\hat{f}(x; \mathcal{D}_{n+1}^y) = \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{1, \ldots, n\}} y_i + \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{n+1\}} y_i = \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{1, \ldots, n\}} y_i + \frac{1}{K} \chi_{\left\{n+1 \in N(x, \mathcal{D}_{n+1}^y)\right\}} y,$$

we see that it suffices to define

$$a(x; \mathcal{D}_n, X_{n+1}) = \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{1, \ldots, n\}} y_i$$

and

$$b(x; \mathcal{D}_n, X_{n+1}) = \frac{1}{K} \chi_{\left\{n+1 \in N(x, \mathcal{D}_{n+1}^y)\right\}}.$$

This completes the proof.////

# Exercise 3

Given data set $(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$ with $\mathcal{Y} = \mathbb{R}$, level $\alpha \in (0, 1)$, we discretize $[-B, B]$, where $B > 0$, by setting, for a positive integer $M$ and all $m \in \{1, \ldots, M\}$:

$$y^{(m)} = -B + \frac{m-1}{M-1} \times 2B.$$

Put, for any $y \in \mathcal{Y}$:

$$\mathcal{D}_{n+1}^{y} = \{(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, y)\}.$$

With a real valued score function $s((x, y); \mathcal{D})$ for $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and $\mathcal{D}$ is any data set containing points from $\mathcal{X} \times \mathcal{Y}$, we consider a discretized conformal prediction set defined as follows. First of all, let $I$ be the set of all $m \in \{1, \ldots, M\}$ such that

$$s\left((X_{n+1}, y^{(m)}); \mathcal{D}_{n+1}^{y^{(m)}}\right) \leqslant \text{Quantile}\left(\left\{s\left((X_i, Y_i); \mathcal{D}_{n+1}^{y^{(m)}}\right)\right\}_{i=1}^{n}; (1-\alpha)\left(1 + \frac{1}{n}\right)\right).$$

Then, put

$$\mathcal{C}(X_{n+1}) = \bigcup_{m \in I} (y^{(m-1)}, y^{(m+1)}).$$

Construct an example of a symmetric score function $s$ and a distribution $P$ of the data set such that

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} = 0$$

for any choice of $B$ and $M$.

**Solution** We construct the example as follows. First of all, map the set $\mathcal{X}$ to $\mathbb{R}$ in any way one wants. From now on, we shall identify $\mathcal{X}$ with $\mathbb{R}$ and we will write for simplicity $\mathcal{X} = \mathbb{R}$. We construct it such that the pairs $(X_i, Y_i)$ ranged over $i \in \{1, \ldots, n+1\}$ are independent identically distributed with distribution of $X_i$'s being any fixed continuous probability measure (say, standard normal) and $Y_i = X_i$. As of the score, we simply put

$$s((x, y); \mathcal{D}) = |x - y|$$

for all $(x, y)$ and all data set $\mathcal{D}$. Clearly, $s$ is symmetric because it does not even depend on $\mathcal{D}$. If $i \in \{1, \ldots, n\}$, our construction yields, almost surely, that

$$s\left((X_i, Y_i); \mathcal{D}_{n+1}^{y^{(m)}}\right) = |X_i - Y_i| = 0$$

and

$$\text{Quantile}\left(\left\{s\left((X_i, Y_i); \mathcal{D}_{n+1}^{y^{(m)}}\right)\right\}_{i=1}^{n}; (1-\alpha)\left(1 + \frac{1}{n}\right)\right) = 0$$

Therefore, $I = \emptyset$ and $\mathcal{C}(X_{n+1}) = \emptyset$ for all choice of $M$ and all choice of $B$. ////

# Exercise 4

For a data set $\mathcal{D} = ((X_1, Y_1), \ldots, (X_m, Y_m)) \in (\mathcal{X} \times \mathcal{Y})^m$, where $\mathcal{Y} = \mathbb{R}$, and any permutation $\sigma \in S_m$, put

$$\mathcal{D}_\sigma = ((X_{\sigma(1)}, Y_{\sigma(1)}), \ldots, (X_{\sigma(m)}, Y_{\sigma(m)})).$$

Given model $\hat{f}(x; \mathcal{D})$, we construct a randomized score function $s((x, y); \mathcal{D}, \xi)$, where $\xi$ is a random variable independent of the data set, as follows. Let $\sigma_\xi$ be an $S_m$-valued random variable depending only on $\xi$ that has the uniform distribution on $S_m$ if $\xi$ has the uniform distribution on $[0, 1]$. Put

$$s((x, y); \mathcal{D}, \xi) = \left| y - \hat{f}(x; \mathcal{D}_{\sigma_\xi}) \right|$$

for all $(x, y)$ and all values of $\xi$. Prove that $s$ is a symmetric score function in the following sense: for every fixed test points $(x_1^{'}, y_1^{'}), \cdots, (x_{m'}^{'}, y_{m'}^{'})$ and every $\tau \in S_m$, the vector

$$\left( s\left((x_1^{'}, y_1^{'}); \mathcal{D}, \xi\right), \ldots, s\left((x_{m'}^{'}, y_{m'}^{'}); \mathcal{D}, \xi\right) \right) \in \mathbb{R}^{m'}$$

has the same distribution as

$$\left( s\left((x_1^{'}, y_1^{'}); \mathcal{D}_\tau, \xi\right), \ldots, s\left((x_{m'}^{'}, y_{m'}^{'}); \mathcal{D}_\tau, \xi\right) \right) \in \mathbb{R}^{m'}$$

when $\xi$ has the uniform distribution on $[0, 1]$.

**Solution** For each $i \in \{1, \ldots, m'\}$, the random variable

$$s\left((x_i^{'}, y_i^{'}); \mathcal{D}, \xi\right) = \left| y_i^{'} - \hat{f}(x_i^{'}; \mathcal{D}_{\sigma_\xi}) \right|$$

has the same distribution as

$$\left| y_i^{'} - \hat{f}(x_i^{'}; \mathcal{D}_{\tau \circ \sigma_\xi}) \right| = s\left((x_i^{'}, y_i^{'}); \mathcal{D}_\tau, \xi\right)$$

as long as $\xi$ has the uniform distribution on $[0, 1]$ because, by construction, both $\sigma_\xi$ and $\tau \circ \sigma_\xi$ have uniform distribution on $S_m$. ////